Who Can Pay for Compliance: Donor Aid Governance and TRIPS-plus Commitments in North-South Trade Agreements*

Jihye Park†

Conditionally accepted at *International Studies Quarterly* as of 10.24.2026

Abstract

Developing countries accepting TRIPS-plus provisions in North-South trade agreements take on concentrated domestic costs, yet the aid that follows signature varies sharply. When does that aid operate as compensation? Prevailing accounts emphasize market-access leverage, leaving donor side payments outside the bargain. I argue that aid operates as compensation only when three conditions hold jointly, two on the donor side and one on the recipient side. The donor's IP demands must be intrusive enough to generate politically salient adjustment costs, and its aid governance centralized enough to tie disbursements to discrete concessions. The recipient's institutions must make the commitment credible to the donor. The United States satisfies both donor-side conditions, the European Union neither, and regime type governs the third. In a PanelMatch design on WTO developing economies, 1995 to 2024, I find that democratic signatories of US agreements receive an immediate increase in US aid of roughly 36% in the signature year, concentrated among recipients with prior major-donor trade experience. The observed autocratic signatories show no comparable increase. EU TRIPS-plus agreements produce no sustained response from either regime type. Donor institutional coherence and the intrusiveness of IP demands sit at the center of TRIPS-plus diffusion.

Keywords: North-South trade agreements, TRIPS, intellectual property rights, foreign aid

*The earlier version of this paper, "Leveraging for Better Investment Grounds", was presented at 2022 IPES Conference. I appreciate the valuable feedback from the panel participants.

†Postdoctoral Researcher, Department of Political Science & International Relations, Université de Genève, Bd du Pont d'Arve 40, Genève, CH-1205. Jihye.Park@unige.ch.

1 Introduction

The World Trade Organization's Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) turned intellectual property protection into an enforceable multilateral obligation, and over the following three decades the United States and the European Union embedded stronger "TRIPS-plus" requirements in their preferential trade agreements (PTAs). These provisions extend patent terms, grant pharmaceutical firms exclusive control over clinical trial data in ways that delay generic entry, restrict compulsory licensing, and impose enforcement obligations that raise compliance costs for developing-country regulators (Fink and Reichenmiller, 2006; Maskus, 2000; Shadlen, 2005). They secure rents for innovation-intensive firms in advanced economies and shift concentrated adjustment costs onto developing-country health systems, generic producers, and consumers of medicines. This asymmetry raises two questions that the literature has tended to run together. The first asks why emerging economies accept obligations that impose concentrated losses on politically visible constituencies. The second asks why, given that countries across the developing world do accept them, the donor aid that follows signature varies so sharply from one recipient to the next. This paper takes up the second and asks under what conditions that aid operates as compensation for the concession. The research design identifies who is compensated given adoption. The decision to adopt lies outside its scope.

Prevailing accounts emphasize the bargaining advantages of large import markets, which the United States and the European Union use to extract deeper commitments from smaller partners, especially under the shadow of unilateral preference withdrawal (Manger and Shadlen, 2014; Dür and Mödlhamer, 2022; Dür et al., 2023; Manger, 2009). These accounts explain why developing countries enter PTAs with major markets and say less about how those countries manage the concentrated domestic costs that intrusive obligations impose. Foreign aid has been hinted at as a possible solution (Baccini and Urpelainen, 2012; Bearce and Tirone, 2010; Bueno de Mesquita and A. Smith, 2007; Bueno de Mesquita and A. Smith, 2009), yet its role as a side payment for intellectual property concessions has not been systematically theorized.

This paper holds market-access leverage as a background scope condition and isolates the conditions under which aid compensates TRIPS-plus concessions. Aid functions as a side payment only when three conditions hold jointly, two on the donor side and one on the recipient side. The donor's IP demands must be intrusive enough to generate politically salient adjustment costs, the

donor's aid governance must be centralized enough to credibly tie disbursements to discrete policy concessions, and the recipient's domestic institutions must make the policy commitment credible to the donor, which in practice favors democratic recipients. Bermeo (2017) and Bermeo (2018) supply the foundation for treating trade and aid as joint instruments of donor strategy, and Bueno de Mesquita and A. Smith (2009) formalize aid as an exchange for policy concessions. Without any one of the three conditions, the compensation channel does not operate.

The two donors at the center of TRIPS-plus diffusion differ on both donor-side margins. The United States drives maximalist TRIPS-plus standards, particularly in pharmaceuticals and enforcement (Sell, 2003; Fink and Reichenmiller, 2006; Drahos and Braithwaite, 2002; Morin, 2009), and concentrates allocation authority over the aid budget in a single executive (Fleck and Kilby, 2010; Milner and Tingley, 2010; Bermeo, 2018). The European Union writes lighter IP chapters (Roffe, 2014; Drexl, 2014; Maskus, 2014) and disburses aid through a fragmented architecture of Commission and member-state budgets (Carbone, 2007; Carbone, 2013; Kim and Jensen, 2018; Dietrich, 2021; Knack and Rahman, 2007). US TRIPS-plus signatures should therefore be followed by discernible aid surges, especially for democratic recipients whose institutions support credible compensation (Carter and Stone, 2015; Milner and Tingley, 2010), while EU TRIPS-plus signatures should not.

Empirically, I construct a panel of self-declared developing economies in the WTO from 1995 to 2024, with TRIPS-plus provisions coded from Morin and Surbeck (2020), US bilateral aid from ForeignAssistance.gov, and EU bilateral aid from the OECD DAC2A. PanelMatch (Imai et al., 2023) with Covariate Balancing Propensity Score weighting estimates the effect of TRIPS-plus signature on subsequent aid flows, separately for democratic and autocratic recipients. Democratic developing countries that sign with the United States receive an immediate increase in US aid of roughly 36% in the year of signature and no further effect in later years. Autocratic signatories of US agreements show no positive surge, and signatories of EU agreements show no sustained positive response from either regime type.

The next section develops the theory. The empirical section introduces the data and design and reports the estimates for both donors and both regime types. The conclusion draws out implications for the diffusion of regulatory standards.

2 A Conditional Theory of Compensatory Aid

TRIPS placed enforceable IP rules at the center of the multilateral trading system and, unlike the earlier World Intellectual Property Organization framework, backed them with binding dispute settlement and retaliatory sanctions (Helfer, 2004). The agreement reflected the preferences of the United States and the European Community, whose firms held most global intellectual property assets. As Sell (2003) documents, pharmaceutical, software, and entertainment multinationals lobbied the United States Trade Representative to pursue high-standard IP protections in both multilateral and preferential arenas. Following TRIPS, the United States consistently incorporated TRIPS-plus obligations into all PTAs negotiated after 1999 (Fink and Reichenmiller, 2006; Drahos and Braithwaite, 2002; Morin and Surbeck, 2020). These obligations include extended patent terms, exclusivity tied to regulatory delays, restricted compulsory licensing, 70 years of copyright protection, and stringent enforcement rules, and they go well beyond WTO requirements.

The distributional consequences of these obligations are well documented. Theoretical models of innovation and technology transfer predict that stronger IP protections channel rents toward innovation-intensive economies and suppress local technological upgrading in developing countries (Maskus, 2000; Glass and Wu, 2007; McCalman, 2001). Ex ante simulation studies estimate that proposed IP reforms would raise drug prices, delay generic entry, and reduce consumer welfare (R. D. Smith et al., 2009). Branstetter et al. (2006) use US firm-level panel data to show that stronger IP rights raise intra-firm technology transfer to multinationals but do not measurably increase broader knowledge diffusion. Duggan et al. (2016) provide quasi-experimental evidence from India that pharmaceutical product patents introduced in compliance with TRIPS raise domestic drug prices without generating offsetting innovation. Shadlen (2005) documents in qualitative detail how developing countries that accepted deep integration commitments in exchange for preferential market access experienced constrained industrial policy space.

The welfare incidence of these provisions is broad. Consumers of patented medicines bear the largest aggregate losses, and in many developing economies few domestic firms compete head-to-head with IP holders of wealthy countries, so the losses fall on diffuse buyers. Political mobilization concentrates among a smaller and better organized set of actors. Generic pharmaceutical producers, health ministries, and patient associations are organized, informed, and capable of threatening ratification or implementation, whereas dispersed consumers rarely organize to comparable effect

(Shadlen et al., 2011; R. D. Smith et al., 2009). Antoine et al. (2024) show that heightened public salience around trade negotiations mobilizes domestic resistance even when external bargaining power is lopsided. Governments contemplating TRIPS-plus signature therefore confront a familiar political economy problem, in which the benefits of preferential access accrue to diffuse actors while the costs fall on smaller and better organized groups whose mobilization can block the deal. Those organized losers generate the political demand that a compensating transfer must satisfy. Peru's trade promotion agreement, taken up below, shows how wide that coalition runs and what the transfer looks like when a donor supplies it.

The compensatory function of foreign aid has long been theorized in the political economy of trade reform. Bueno de Mesquita and A. Smith (2009) formalize aid as an exchange in which donors purchase policy concessions from recipient leaders, who in turn use the resources to compensate or buy off potential losers. Bueno de Mesquita and A. Smith (2007) and Bearce and Tirone (2010) provide complementary evidence on aid-for-policy bargains, and Baccini and Urpelainen (2012) demonstrate that aid assists governments in adopting PTAs that would otherwise provoke destabilizing opposition. Bermeo (2017) and Bermeo (2018) integrate these insights into a broader account of targeted development, in which trade and aid are joint instruments donors use to pursue policy objectives in countries with developmental potential. The TRIPS-plus case fits naturally within this framework, since the provisions impose concentrated costs on identifiable constituencies and the donor has a strong stake in adoption and implementation. These formulations share a fiscal assumption. The transfer is fungible revenue that the recipient leader receives and reallocates, and the recipient's credibility consists in refraining from diverting it. The mechanism developed below does not carry that assumption, because a transfer offsets the political cost of a concession whenever it reaches the affected constituencies, whatever route it takes to reach them. Whether aid actually operates as compensation, however, depends on conditions that hold for some donors and not others.

The compensatory channel operates only when three conditions hold together. The first is a demand condition. The donor's IP demands must be intrusive enough to impose politically salient adjustment costs on organized domestic constituencies, so that the recipient government has a genuine need for those costs to be offset. Where the obligations are routine or readily absorbed within existing regulatory frameworks, that need is weak and no compensation is required. The second is a supply condition. The donor's aid apparatus must concentrate allocation authority in a

principal that can tie a disbursement to a discrete policy concession, so that the donor can deliver a targeted transfer at the moment of commitment. The third is a credibility condition on the recipient side. The recipient's domestic institutions must make the policy commitment credible to the donor, which in practice favors democratic recipients (Martin, 2000; Mansfield et al., 2000). The three conditions enter conjunctively. Each is necessary and none is sufficient on its own, so a recipient that needs compensation cannot obtain it from a donor unable to supply a targeted transfer, a donor able to supply one has no occasion to when its demands impose no politically costly adjustment, and a donor and recipient aligned on demand and supply still produce no durable exchange when the recipient cannot credibly implement what it signs.

Because the conditions are jointly necessary, the theory makes predictions for donors beyond the two examined here. Consider a donor whose IP demands are intrusive but whose allocation authority is split among rival principals. Japan is the closest case. Its economic partnership agreements carry substantial IP obligations in the Morin and Surbeck (2020) coding, yet its aid budget is parcelled among ministries that each hold distinct budgets and priorities. The Ministry of Foreign Affairs, the Ministry of Finance, and the Ministry of Economy, Trade and Industry each command their own ODA allocations, and no executive gathers these into a single hand (Arase, 1995; Lancaster, 2006). The barrier to compensation lies in that division of budget authority, above the Japan International Cooperation Agency that implements much of the assistance. A recipient that accepts a Japanese agreement, as Indonesia did in 2007, generates a demand for compensation that no Japanese principal can time to its signature, and no compensatory response should follow. A donor with concentrated aid authority but modest IP demands presents the reverse combination. It can supply a targeted transfer but faces little recipient demand for one, and again no response should follow. The compensatory pattern emerges only when intrusive demands and concentrated allocation authority coincide and the recipient can credibly implement.

This structure treats market-access leverage as a background scope condition. The bargaining asymmetries that bring developing countries to the table with large markets are common to all of these donors and cannot by themselves account for the divergent aid response. Market size determines whether a TRIPS-plus agreement is negotiated at all, and the three conditions determine whether aid is paid around its signature.

Compensation is one of several mechanisms that could link a TRIPS-plus signature to a donor's aid, and the argument is stated against these rivals. An emulation account would attribute

IP adoption to the influence of peer adopters and would give donor aid no particular role (Simmons et al., 2006). A coercion account would expect donors to wield the threat of aid withdrawal (Mangini, 2026; Farrell and Newman, 2019). A geopolitical-reward account would tie transfers to a recipient's strategic alignment with the donor, and would extend to autocratic partners as readily as democratic ones (Alesina and Dollar, 2000; Dreher et al., 2009). A bundled-side-payment account would trace any transfer to the trade agreement as a bundle, with the IP chapter carrying no separate weight (Baccini and Urpelainen, 2012). What distinguishes the compensation account from these rivals is its conjunctive structure. It expects transfers only where an intrusive IP demand, a donor able to supply a targeted payment, and a credible recipient coincide, a combination that none of the alternative mechanisms requires.

The United States satisfies the first two conditions, and the case for treating it as the compensating donor begins with the intrusiveness of its demands. A large body of scholarship identifies the United States as the central global driver of maximalist TRIPS-plus standards, particularly in pharmaceuticals and enforcement (Sell, 2003; Fink and Reichenmiller, 2006; Drahos and Braithwaite, 2002). US PTAs typically contain extensive data exclusivity, restrictions on compulsory licensing, enhanced enforcement rules, patent term extensions, and patent linkage, which compels regulators to notify patent holders before approving generic competitors, and together these delay generic entry and constrain regulatory flexibility far beyond the TRIPS baseline. Morin (2009) shows that US bilateral commitments set new baselines that then diffuse through multilateral forums, and Lindstrom (2010) and Turk (2010) document the take-it-or-leave-it posture the United States adopts in IP chapters. Table A.1 in the Appendix summarizes this substantive content alongside the lighter EU alternative. These demands generate the concentrated adjustment costs that create recipient demand for compensation, and intrusiveness enters the argument as a cross-donor contrast. The willingness to supply compensation follows from the scale of the rents at stake. The IP-intensive industries that lobby for these standards capture monopoly rents that dwarf the aid outlays involved in easing a partner's compliance, and Sell (2010) documents how PhRMA and allied associations drove successive escalations in USTR demands across administrations. A targeted transfer measured in tens of millions of dollars is a modest price for protections worth far more to the donor's firms.

The United States also satisfies the supply condition, though the term centralization requires care. On one common meaning the United States does not operate a centralized aid apparatus

at all. Its assistance is administered by many agencies, including USAID, the State Department, the Millennium Challenge Corporation, the Departments of Agriculture and Defense, and others, and by that measure US aid is institutionally dispersed (Lawson, 2016). The dimension that matters for compensation is allocation authority. A single executive, accountable to a single national legislature, can direct the aid budget toward a chosen policy objective and time a disbursement to coincide with a treaty commitment, whatever the number of implementing agencies through which the money ultimately flows. What the United States possesses is concentrated allocation authority under executive direction (Fleck and Kilby, 2010; Milner and Tingley, 2010). Bermeo (2018) and Bermeo (2017) show that the United States calibrates assistance toward countries undertaking reforms aligned with its commercial and geopolitical interests.

The compensatory transfer concentrates at the moment of commitment, and the timing follows from a commitment problem. Before a PTA is signed, the domestic costs of TRIPS-plus obligations remain prospective, opposition has not fully mobilized, and the government retains the option to walk away, so resources transferred in advance risk being wasted if negotiations collapse or the partner refuses the IP chapter. Signature locks in the recipient's regulatory commitment and signals to the donor that compensation will be met with compliance. The political demand for offsetting transfers therefore peaks at or immediately after signature, when governments must manage the concentrated losses of organized constituencies as implementation looms and donors can link disbursements to an observable and irreversible policy act (Baccini and Urpelainen, 2012; Bueno de Mesquita and A. Smith, 2007). In the terms of the aid-conditionality literature the mechanism is transactional and ex post, with disbursements following discrete policy acts, whereas threshold conditionality of the Millennium Challenge Account type requires eligibility before aid flows (Öhler et al., 2012; Kilby, 2009). Dreher et al. (2008) show that US aid allocation responds to specific recipient policy behaviors, which is the allocation logic a transactional side payment relies on.

One caveat attaches to this timing logic. Public health advocacy and generic-industry lobbying can mobilize opposition well before formal commitment, so some demand for offsetting resources may arise earlier than the signature moment, and some technical cooperation for patent offices and regulatory agencies does flow during negotiation. That pre-signature assistance addresses institutional readiness for implementing the new obligations.

Peru illustrates the demand condition and the form the corresponding transfer takes. Peru

signed its trade promotion agreement with the United States in April 2006 and ranks among the largest recipients of US trade capacity building assistance in the years surrounding that signature, standing twenty-first among all recipients in FY2007 (Langton, 2008). US trade capacity building obligations to Peru stood at 10.9 million dollars in FY2005, rose to 32.4 million in FY2006, the fiscal year containing the signature, and fell back to 8.8 million in FY2007. Trade capacity building is a category of aid tracked apart from general development programming and defined by its trade-related purpose. It is delivered through areas that run to customs operation and administration, trade facilitation, export promotion, trade-related agricultural development, human resources and labor standards, environmental sector trade and standards, and competition policy and foreign investment. The list contains no area for intellectual property, and none for pharmaceuticals or health. The trade capacity building committee established under the agreement was directed toward small and medium-sized enterprises, rural farmers, transport, and telecommunications (Office of the United States Trade Representative, 2005).

The breadth of the affected coalition is the demand condition in concrete form, and the account given by Peru's own head negotiator for the intellectual property chapter shows how wide that coalition ran (García, 2008). The Ministry of Commerce and Tourism accredited 131 institutions to the negotiation, among them twenty-three regional governments, twelve ministries, forty management unions, and three work unions. The Ministry of Health took part directly in the intellectual property table and its minister joined the final two rounds. The national generic producers, organized as ADIFAN, published a ten-point objection in July 2005 that asked for intellectual property to be withdrawn from the negotiation altogether and opposed test-data exclusivity, patent term extension, and restrictions on compulsory licensing. ForoSalud, a civil society forum on public health, tied delayed generic entry to patients under treatment for tuberculosis, HIV, diabetes, hypertension, and cancer. An intrusive intellectual property chapter raises the political price of the agreement as a whole, and a transfer that answers that price is addressed to a coalition of this kind. Dietrich (2013) documents that US assistance frequently arrives through non-state implementers and bypasses recipient budgets.

The case is offered as illustration and warrants a cautious reading, as the record does not disclose which constituencies receive the transfer. It establishes that an organized coalition of prospective losers formed around the intellectual property chapter, that its membership ran well past the pharmaceutical sector, and that trade-related assistance showed a stark surge in the

signature year. It does not trace the transfer to those constituencies, and the negotiator's own assessment is that ADIFAN and ForoSalud were widely regarded as advancing a political agenda, which reduced their influence over the outcome. The three impact studies commissioned in Peru disagreed with one another about whether the chapter would raise medicine prices at all. Peru's domestic politics of the intellectual property chapter are legible in the record, and the incidence of the assistance that followed is not.

The recipient side supplies the third condition, and it explains why the compensatory channel should not operate uniformly even within the universe of US TRIPS-plus signatories. Two rationales are at play. First, donors require credible implementation. Democratic recipients face stronger institutional incentives to follow through on the commitments they sign, because their leaders are constrained by electoral accountability and legislative oversight that raise the cost of abandoning ratified commitments (Baccini and Urpelainen, 2014; Martin, 2000; Mansfield et al., 2000). The canonical accounts locate recipient credibility elsewhere, in the recipient government's forbearance from diverting fungible aid (Bueno de Mesquita and A. Smith, 2009; Baccini and Urpelainen, 2012). That version of the credibility condition presumes the transfer enters the recipient budget. The condition invoked here concerns the durability of the policy commitment and holds whoever has custody of the funds.

Second, donor-side rules of aid allocation systematically advantage democracies. Carter and Stone (2015) locate the binding constraint on US aid in political support within Congress for aid to particular countries, with the overall size of the aid budget a looser constraint. Congress has built a detailed appropriation process through which members shape commitments while delegating discretion over disbursements to the executive, and providing aid to autocratic governments is controversial enough that the executive looks first to democracies when it seeks to purchase a policy concession. An aid linkage to a discrete concession is therefore credible when directed toward a democratic recipient and costly to sustain when directed toward an autocratic one. Milner and Tingley (2010) provide complementary evidence on the legislative politics of US aid. More broadly, Alesina and Dollar (2000) find that donors reward recipient democracy and democratization with larger aid allocations. The compensatory response should therefore appear most clearly for democratic signatories of US PTAs.

Implementation credibility also has a pre-history component. Recipients that have already navigated formal trade commitments with major OECD donors carry an observable record

of regulatory delivery that makes targeted disbursements safer to extend, while recipients whose first major-donor PTA is the very agreement under negotiation present a thinner credibility signal. The reputational logic is familiar from work on compliance with formal international commitments, where a government's record of honoring past obligations informs how counterparts assess its future ones (Simmons, 2000). It meets a donor side already disposed to read such records, since aid allocation has grown more selective on observable institutional performance across the period studied here (Dollar and Levin, 2006).

H1. Democratic developing countries that sign TRIPS-plus agreements with the United States receive larger volumes of US bilateral aid.

The European Union sits below the United States on both donor-side margins, and the difference begins with intrusiveness. EU agreements do carry obligations that exceed the TRIPS baseline, so the contrast drawn here is between a heavier and a lighter TRIPS-plus package. Comparative analyses confirm that the European Union takes a more flexible approach in its IP chapters, one that accommodates the regulatory traditions of partner countries (Roffe, 2014; Seuba, 2014; Maskus, 2014; Drexl, 2014). EU Economic Partnership Agreements with African, Caribbean, and Pacific states include IP provisions but typically allow longer transition periods and preserve greater policy space than comparable US agreements. Where the EU does pursue ambitious TRIPS-plus standards, its priorities differ. EU agreements emphasize geographical indication protection for agricultural products and longer data exclusivity periods under the $8 + 2 + 1$ framework, and they place less weight on the patent linkage and enforcement mechanisms that most directly constrain generic pharmaceutical entry. The political adjustment cost of accepting an EU package is therefore smaller than the cost of a US package, and recipient demand for compensation is correspondingly weaker.

EU allocation authority is dispersed in the same measure. Carbone (2007) documents that EU development policy operates through an architecture in which the European Commission and 27 member states maintain distinct aid budgets, strategic priorities, and implementation channels. Carbone (2013) characterizes this as competing logics of EU aid, with a developmental logic remaining the more institutionally entrenched, and Carbone (2012) shows that coherence between EU trade and development objectives remains an aspiration without institutional guarantee. Kim and Jensen (2018) confirm empirically that EU aid patterns reflect dispersed preferences among multiple principals, and Dietrich (2021) shows more generally that donor institutional structures

systematically shape whether aid is used to pursue policy objectives, with developmental logics tending to dominate where authority is dispersed across many principals. The dispersion of EU allocation authority tilts its aid toward long-running developmental programming. Because the European Union satisfies neither the demand nor the supply condition, its aid should not respond to TRIPS-plus signature, and because these are donor-side failures the prediction holds regardless of recipient regime type.

H2. Signatories of TRIPS-plus agreements with the European Union receive no comparable compensatory increases in EU bilateral aid, regardless of regime type.

3 Empirical evidence

3.1 Data and Identification

The sample consists of self-declared developing economies in the WTO observed annually from 1995 to 2024. Self-declared WTO status determines eligibility for the concessional trade schemes within which TRIPS-plus PTAs are negotiated.

The treatment of interest is the signature of a TRIPS-plus PTA. I draw the binary indicator from the TRIPS-plus dataset of Morin and Surbeck (2020), which codes whether each PTA between a major developed economy and a developing country contains any TRIPS-plus provisions. The dataset also codes depth across intellectual property domains, but the binary indicator suffices here because the question is whether signature triggers a compensatory response, and the near-uniform maximalism of US IP chapters leaves little within-US depth variation to exploit.

I date treatment to the calendar year in which a TRIPS-plus PTA is signed. Signature is the point of commitment the theory identifies, when the regulatory concession becomes public and irreversible and the donor can attach a transfer to it. Negotiation opens years earlier and ratification often follows years later, but signature is the moment at which the bilateral bargain is struck. Annual observation cannot distinguish an agreement signed in January from one signed in December, and the pre-treatment placebos below show that aid does not begin rising before the signature year.

The outcome is bilateral aid disbursements from each donor to each recipient. The analysis

uses a separate data source for each donor, which mirrors the institutional asymmetry at the core of the theoretical argument.[1]

The estimator is PanelMatch (Imai et al., 2023). PanelMatch suits settings in which treatment adoption is staggered, non-random, and shaped by prior characteristics that also affect the outcome, as TRIPS-plus participation is. It constructs matched sets of treated and control units with identical treatment histories and similar covariate profiles before treatment, refined with Covariate Balancing Propensity Score (CBPS) weighting on the covariates described below.

To test the regime-conditioning prediction, I estimate the treatment effect separately for democratic and autocratic recipients. Including democracy as a moderator inside a single estimator would conflate regime status with treatment status, so I use two parallel within-regime treatments. The within-democracy treatment indicator equals one in country-years in which a democratic developing country has signed a TRIPS-plus PTA with the donor and zero otherwise, and the within-autocracy indicator is defined analogously. Both arms use the same lag-lead structure and the same CBPS-weighted refinement, producing two ATT trajectories that are directly comparable across regime types.

Democracy is constructed from the Polity V dataset (Marshall and Gurr, 2020), with Polity scores above zero coded as democratic. The threshold does not drive the arms that carry inferential weight. Re-estimating all four arms under the stricter conventional Polity cutoff of 6, and again under a V-Dem electoral-democracy cutoff of 0.5 (Coppedge et al., 2026), leaves the within-democracy US estimate numerically identical, because every democratic US signatory clears each of these thresholds in its signature year. The within-democracy EU estimate stays negative and statistically insignificant under all three codings. The one cell that moves is the within-autocracy EU arm, whose $t + 0$ estimate reaches statistical significance under the V-Dem coding on nine treated country-years, and that arm is reported below as a descriptive comparison for this reason. Table A.10 in the Appendix reports the full set. The choice of zero nonetheless follows from what the theory asks of recipient institutions. Both rationales for the regime distinction turn on whether the executive faces institutional constraints that make a policy commitment durable and whether the recipient is a politically appropriable target for congressional appropriations, and both operate

[1] US disbursements come from the ForeignAssistance.gov database maintained by USAID, which records US official development assistance at the activity level with timing detail that DAC aggregates lack (Fleck and Kilby, 2010; Bermeo, 2018). EU disbursements come from the OECD DAC2A, which aggregates Commission and member-state flows, since no single-source database exists for a donor whose aid authority is divided across the Commission and 27 member states (Carbone, 2007; Kim and Jensen, 2018).

at the boundary between constrained and unconstrained executives. In a developing-country sample a cutoff of 6 places partial democracies that have meaningful executive constraints on the autocratic side of that boundary, which mixes two institutional types inside the autocracy arm and blunts the contrast the design is built to draw. The lag is 4 years and the lead window is 0 through 3 years after signature, symmetric around the moment of treatment. Inference uses bootstrapped intervals.

The within-democracy US arm rests on 9 treated countries, the within-autocracy US arm on 3, the within-democracy EU arm on 31, and the within-autocracy EU arm on 3, with the full counts of treated units and matched control sets in Table A.9 in the Appendix. With three treated countries an arm is exposed to case composition as much as to statistical precision, since a single unrepresentative signatory can move the estimate and no amount of additional bootstrap resampling would reveal it. The two autocracy arms are therefore reported as descriptive comparisons, and they carry less evidentiary weight than the democracy arms. The US autocracy arm comprises Bahrain, Morocco, and Oman, and the EU autocracy arm Haiti, Peru, and Eswatini, so readers can judge how far these cases travel. The nulls in these arms speak to the observed signatories and leave open how autocratic recipients as a class would respond.

The identifying assumption is sequential ignorability conditional on the matched set. Treatment is as good as random within the pool of country-years matched on prior treatment history, lagged outcomes, and CBPS-weighted covariates (Imai et al., 2023). Donors choose which developing countries to approach based on strategic alignment, market size, regime trajectory, prior trade volume, and the existing aid relationship. Recipient governments choose whether to accept the IP chapter based on internal coalitions, lobbying intensity, and the value of preferential access. Both sides may react to common temporal shocks such as US Trade Promotion Authority renewal cycles, donor administration changes, and global business-cycle conditions. PanelMatch addresses each of these channels through a distinct component of its construction.

The matching procedure builds the control pool for each treated country-year from non-treated country-years with identical treatment histories over the prior four years, so a country that signs in year t is compared only to countries that have not yet signed with the same donor as of year t. The CBPS-weighted refinement then balances the matched controls on lagged outcomes over the three preceding years, together with lagged GDP, GDP per capita, and GDP growth, the macroeconomic variables most commonly invoked in donor allocation models (Alesina and Dollar, 2000; Fleck and Kilby, 2010; Bermeo, 2018). Balancing on the lagged outcome answers the

selection concern directly, since donors selecting partners on grounds correlated with future aid leave fingerprints on the pre-treatment aid path, which the balancing holds fixed.

Two further sources of confounding are absorbed by the structure of the comparison. Time-invariant unobservables, including geography, colonial history, language, and baseline aid eligibility under congressional mandates, drop out of the within-unit comparison. Common temporal shocks drop out because treated and control country-years are drawn from the same calendar year, so domestic political and administrative shifts, as well as macroeconomic fluctuations cannot drive the ATT. The residual class of confounders the design cannot absorb consists of time-varying unobservables that shift both treatment and outcome and are uncorrelated with the three-year pre-treatment aid trajectory and the matched covariates. Many such confounders would leave a trace in the pre-treatment periods, which the placebo estimator below tests directly. A contemporaneous shock that moved both the timing of the agreement and aid without disturbing the pre-treatment trajectory would not be detected. The placebo window reaches back to $t-4$, so a donor that began raising aid to prospective signatories more than four years before signature would produce anticipation outside the tested range, and the placebo cannot rule that longer horizon out.

3.2 TRIPS-Plus and Compensatory Aid

Figure 1 reports the PanelMatch estimates for the four arms of the design. Each panel shows the ATT trajectory at $t+0$ through $t+3$ for one combination of donor and recipient regime. Filled markers denote estimates that are statistically significant, hollow markers denote those that are not, and vertical bars show the bootstrapped intervals. Table A.4 in the Appendix reports the full estimates, bootstrapped standard errors, and the number of treated and matched control units for each arm.

The pattern matches H1 closely. Democratic developing countries that sign TRIPS-plus agreements with the United States experience an immediate and statistically significant increase in US aid in the year of signature. The point estimate at $t+0$ corresponds to an increase of 36% in US aid relative to matched control democracies that did not sign ($\exp(0.31)-1 \approx 0.36$). ATT estimates at $t+1$ through $t+3$ are statistically indistinguishable from zero, so the transfer arrives at the moment of commitment and does not recur, which is the timing a compensatory side payment would take and the front-loaded profile that Baccini and Urpelainen (2012) document for trade

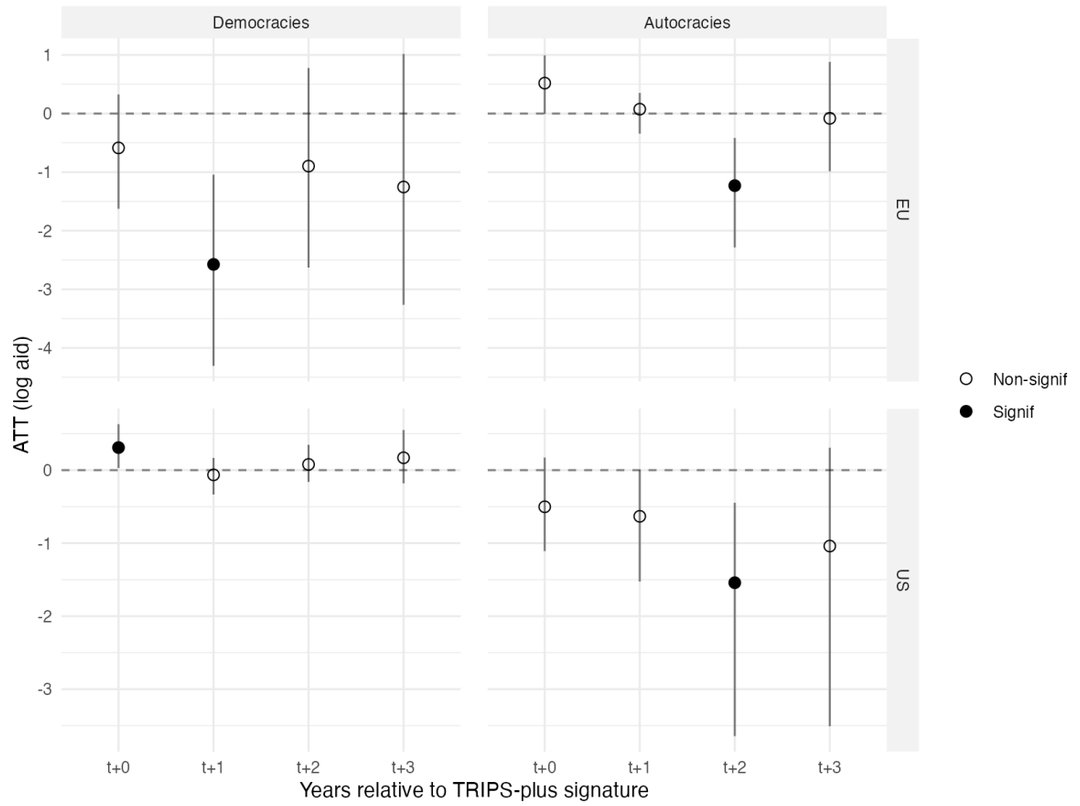


Figure 1: Aid disbursement and TRIPS-plus signature: PanelMatch ATTs by donor and recipient regime type.

Alt-text. A four-panel chart of ATT trajectories from $t+0$ to $t+3$. The US within-democracy panel shows a positive significant estimate at $t+0$ that falls to zero by $t+1$. The US within-autocracy, EU within-democracy, and EU within-autocracy panels show no sustained positive response. Filled markers are significant, hollow markers are not, and vertical bars are bootstrapped intervals.

reforms more generally.[2] A single-year pulse of this kind is consistent both with a forward-looking side payment and with a one-time political signal sent at the moment of agreement, and the design does not adjudicate between the two readings. The thirty-six percent increase is an average effect across a heterogeneous pool of nine democracies. A leave-one-out analysis shows that no single country drives it, with the $t+0$ estimate remaining between 0.23 and 0.43 as each treated country is dropped in turn. The effect concentrates among the subset of recipients with prior major-donor trade experience, and jointly excluding the five reform-credential democracies reduces the pooled $t+0$ estimate to a statistically insignificant 0.08. Table A.11 in the Appendix reports both exercises. The split by prior PTA experience below returns to this concentration, which the theory anticipates through the implementation-credibility condition.

[2]The mechanism requires that compensation reach the constituencies bearing the adjustment costs and carries no requirement about the fiscal conduit. Dietrich (2013) shows that donors engage the recipient government where governance is stronger and bypass it through non-state implementers where it is weak, so her account expects engagement with the established democracies studied here. The data fall short of confirming this. Every ForeignAssistance.gov activity record from FY2001 onward names an implementing partner, but for all Defense and 98% of State disbursements to the treated democracies the recorded partner is the disbursing agency itself, which leaves the downstream conduit unrecoverable. Foreign governments account for 2.4% of signature-year disbursements and US agencies recorded as their own implementers for a further 71%, so the recipient-government share lies somewhere between 2% and 73%. The OECD channel-of-delivery code enters the CRS in 2004 and covers the treated signatures only in part.

The within-autocracy arm of the US case shows the regime contrast the theory predicts. Among the observed autocratic signatories of US agreements there is no evidence of a comparable signature-year increase. The ATT at $t+0$ is negative and statistically insignificant, point estimates at later leads remain negative, and the estimate at $t+2$ is significantly negative. The arm rests on Bahrain, Morocco, and Oman, so it is reported as a descriptive comparison, consistent with the credibility condition without establishing that autocratic recipients as a class are never compensated (Carter and Stone, 2015; Milner and Tingley, 2010). The placebo below shows that these recipients were already on a statistically significant downward US-aid trajectory at $t-4$ and $t-3$, so bilateral aid to the two groups follows different paths before either signs anything. Park (2026) finds the same regime asymmetry across US TRIPS-plus commitments more broadly.

The two EU panels speak to H2. The within-democracy arm shows no positive aid response at $t+0$, with a negative and statistically insignificant point estimate in the year of signature and a negative and statistically significant estimate at $t+1$. The within-autocracy arm shows a positive point estimate at $t+0$ that does not reach statistical significance and a statistically significant negative effect at $t+2$. This EU autocracy arm rests on three treated countries, and the placebo below shows that its $t+0$ point estimate continues an upward trajectory already statistically significant at $t-4$, so the signature year adds nothing to a movement that predates it. Signing a TRIPS-plus agreement with the European Union does not trigger a sustained compensatory aid surge, which is what the theory expects when obligations are less intrusive and aid governance is too dispersed to coordinate transfers around a discrete concession (Carbone, 2007; Kim and Jensen, 2018; Dietrich, 2021; Knack and Rahman, 2007).

Together the four panels are consistent with the joint conditions the theory sets out. The design varies recipient credibility independently, through the democracy and autocracy arms within each donor. It does not vary the two donor-side conditions independently, because the United States both makes the more intrusive demands and holds the more concentrated allocation authority, while the European Union sits below it on both margins at once. The donor contrast therefore supports the conjunctive account without attributing the difference to intrusiveness or to allocation authority separately, or to their interaction. Separating them would require a donor that combines intrusive demands with fragmented authority, of which Japan is the closest case discussed above, or concentrated authority with modest demands, and the sample contains neither. What the four panels do show is that the compensatory pattern appears only where an intrusive and

institutionally concentrated donor meets a credible recipient.

The four arms also bear on what the transfer compensates. A deep agreement with a major economy concentrates adjustment costs on many groups at once, among them import-competing agriculture, textile producers, and firms sheltered by government procurement rules, and each of them could in principle attract a side payment. That exposure varies little across the comparisons drawn here. European Union agreements reach comparable regulatory depth and impose agricultural and services adjustment on developing partners on a similar scale, and signatories of either regime type receive no compensating surge. Autocratic signatories of US agreements accept the same template and carry the same array of injured sectors, and their point estimates run negative. What does vary across the four arms is the intrusiveness of the donor's IP demands and the regime type of the recipient, and the aid response follows those two margins. The estimate is best read as the aid response to a package whose intrusiveness the TRIPS-plus content indexes, and it assigns no fixed share of the increment to any single injured constituency.

3.3 Identification checks and robustness

Pre-treatment placebo. If aid to eventual signatories were already rising in the pre-treatment years, the $t+0$ estimate could not be read as a response to the commitment itself. PanelMatch's `placebo_test` holds the matched sets fixed and recomputes ATTs at $t-4$, $t-3$, and $t-2$ (Imai et al., 2023), reported in Figure 2 and Table A.5 in the Appendix. No pre-treatment ATT in either democracy arm is distinguishable from zero, with US point estimates between -0.25 and -0.16 in log aid units.

US autocracies that eventually sign were on a statistically significant downward US-aid trajectory at $t-4$ and $t-3$ that has flattened to -0.068 by $t-2$. EU autocracies were on a statistically significant upward EU-aid trajectory at $t-4$ that likewise fades by $t-2$, so the positive ATT at $t+0$ for that arm continues a movement already underway. Both autocracy arms therefore depart substantially from their matched controls four years before signature and converge by the eve of it, while both democracy arms hold flat throughout the entire pre-treatment window.

As Section 2 set out, autocracies receive US aid only when they matter to American foreign policy, and that aid is tied to long-term objectives such as regional stability and basing rights (Carter and Stone, 2015; Milner and Tingley, 2010). The volatile pre-treatment path and the

null at signature are both consistent with aid to these recipients answering to a schedule of its own, and three treated countries in each arm cannot establish that the channel is closed to autocratic recipients as a general matter.

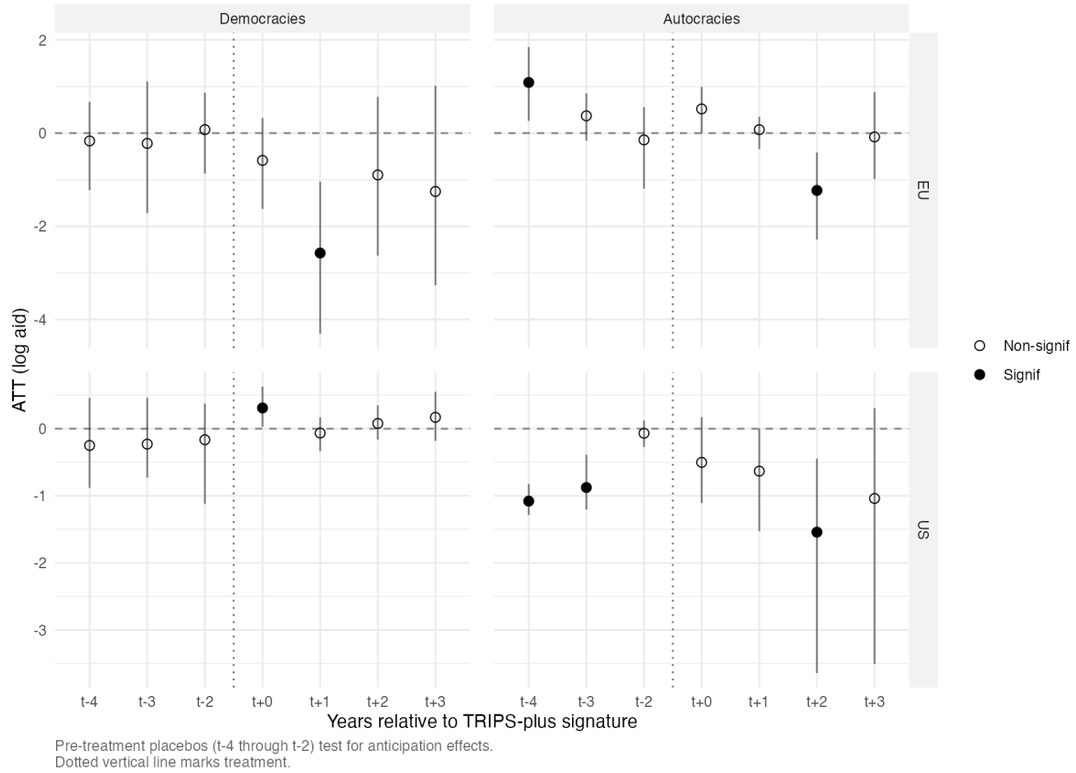


Figure 2: Aid disbursement and TRIPS-plus signature, extended with pre-treatment placebos. Pre-treatment ATTs at $t-4$ through $t-2$ test for anticipation effects. The dotted vertical line marks treatment.

Alt-text. The same four panels extended leftward with pre-treatment placebo estimates at $t-4$, $t-3$, and $t-2$. The US and EU within-democracy arms show flat pre-treatment estimates near zero. A dotted vertical line marks the treatment year.

Recipient credibility and the concentration of the surge. The credibility condition predicts which recipients are paid. A donor extending a targeted transfer around a signature bears the risk that the recipient does not implement what it signed, and a recipient that has already carried a formal trade commitment with a major OECD donor has a record on precisely that risk. The surge should therefore concentrate among experienced recipients.

I split the US within-democracy treated cell by whether the recipient had a prior PTA with any major OECD donor, coded from the Design of Trade Agreements database (Dür et al., 2014), and re-estimate each subsample. Figure 3 reports the result, with Table A.6 in the Appendix giving the estimates. The surge concentrates entirely among recipients with prior major-donor experience (Chile, Colombia, Costa Rica, the Dominican Republic, and Peru), whose $t+0$ ATT is 0.59 in log aid units and statistically significant, corresponding to roughly an eighty percent rise. The without-prior subsample (El Salvador, Guatemala, Honduras, and Panama) shows a small and

statistically indistinguishable estimate of 0.09.

These five countries are the same ones whose joint exclusion reduces the pooled estimate to 0.08 in the influence analysis reported above, so the two exercises describe a single concentration. What the split establishes is that the surge tracks recipient credibility in the way the theory predicts. It does not isolate the IP chapter, since no US PTA in the sample lacks TRIPS-plus content. Prior-PTA recipients also differ from newcomers in income, administrative capacity, and the size of the existing aid relationship as well as in credibility, so the split locates the concentration without isolating its cause.

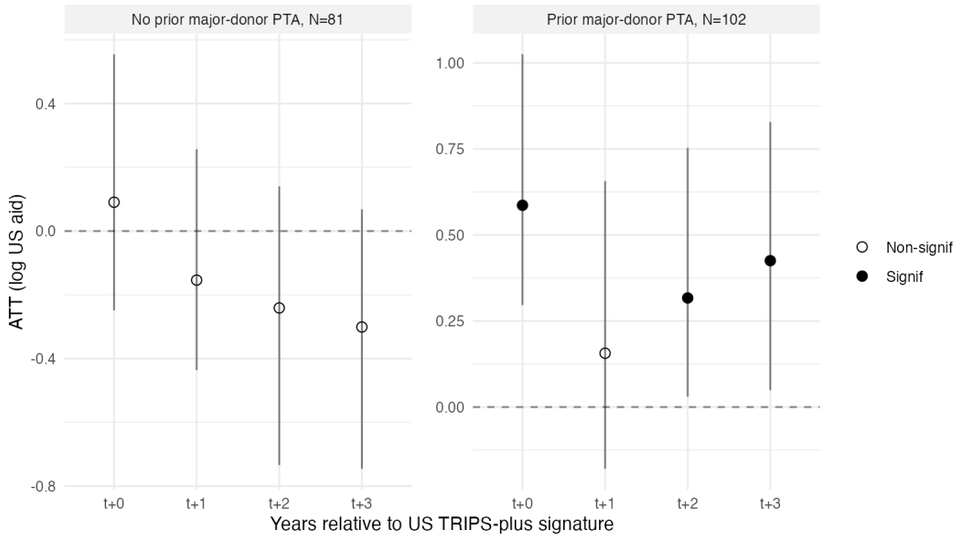


Figure 3: US aid response to TRIPS-plus signature, within-democracy estimates split by whether the recipient had a prior PTA with any major OECD donor.

Alt-text. Two panels of US within-democracy aid ATTs, split by prior major-donor PTA experience. The with-prior group shows a large positive significant $t+0$ estimate, and the without-prior group shows a small estimate indistinguishable from zero.

Non-TRIPS-plus PTA placebo for the EU. To tell a general EU reluctance to couple aid with trade agreements from a response specific to TRIPS-plus content, I re-estimate the EU arm with treatment set to PTAs that lack TRIPS-plus provisions. The dataset of Morin and Surbeck (2020) identifies eight such country-years, mostly Economic Partnership Agreements concluded with African, Caribbean, and Pacific states between 2008 and 2009, together with the early Chile-EC Association Agreement of 2002.

Figure 4 reports the result, with Table A.7 in the Appendix giving the estimates. EU aid responds positively and persistently to non-TRIPS-plus PTA signatures, with pooled ATTs of $+1.66$ at $t+1$, $+1.16$ at $t+2$, and $+1.30$ at $t+3$, all statistically significant. EU aid is therefore responsive to PTA signatures as such, so the headline EU null is no artifact of the EU declining to pair aid with trade agreements. The check also reveals the channel through which EU aid flows, since

the eight signatures sit within the Cotonou Partnership framework that bundles trade preferences with development assistance for ACP states.

EU aid rises around these signings because the agreements are part of an integrated development relationship, whereas EU TRIPS-plus PTAs involve partners in Latin America, the Caribbean, Eastern Europe, and the Mediterranean outside the Cotonou bundling. Because the comparison holds the donor fixed and varies only the TRIPS-plus content of the agreement, it also partly answers the concern that US and EU partner populations simply differ.

The partner populations behind the two donor arms nonetheless differ, which is the most consequential limitation of the comparison, since the United States signed predominantly with Latin American democracies carrying established reform records and no within-arm matching can balance populations that do not overlap. Restricting attention to the eight countries that signed TRIPS-plus PTAs with both donors, all democracies, the US within-democracy $t + 0$ estimate remains positive and close to the full-sample magnitude while the EU estimate remains non-positive, so the donor divergence survives restriction to a shared partner pool, though on a thin base and with the autocracy comparison unidentified.

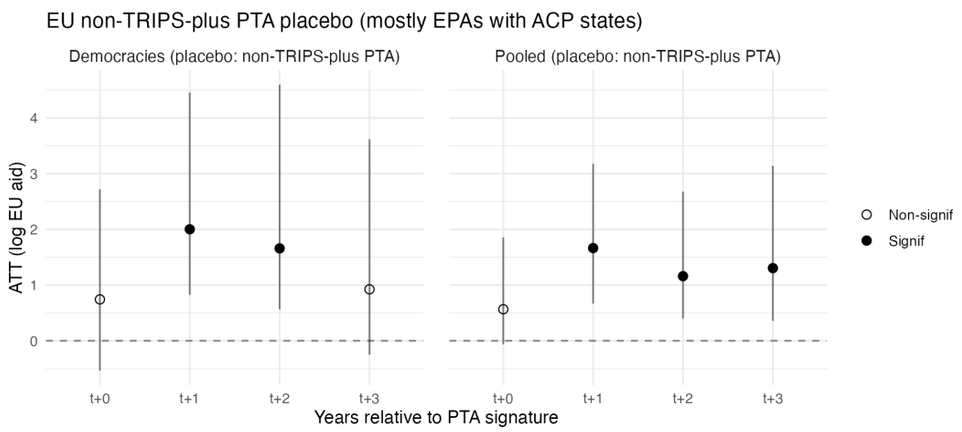


Figure 4: EU aid response to PTA signatures that do not contain TRIPS-plus provisions, mostly EPAs with ACP states.

Alt-text. A single panel of EU aid ATTs following EU PTA signatures that lack TRIPS-plus provisions. Estimates are positive and significant at $t + 1$ through $t + 3$, in contrast to the null EU response to TRIPS-plus signatures.

MFN sequencing in EU TRIPS-plus signatures. The marginal cost of accepting EU TRIPS-plus provisions depends on whether the recipient already holds a US TRIPS-plus PTA. The GATT exception to most-favored-nation treatment under Article XXIV applies to tariff preferences and does not extend to TRIPS, which is non-discriminatory by design. A country that has already accepted US TRIPS-plus obligations therefore faces lower marginal adjustment when signing an EU

TRIPS-plus PTA, so the EU response should be more muted in the with-prior-US sub-population.

Splitting the EU within-democracy arm by prior US TRIPS-plus status (Figure 5, with Table A.8 in the Appendix), both sub-samples show non-positive responses, and the with-prior-US group shows somewhat smaller absolute coefficients ($t+1$ ATT of -2.01 versus -2.94), in the predicted direction though not statistically distinguishable. The EU null is therefore not confined to recipients that already hold US obligations.

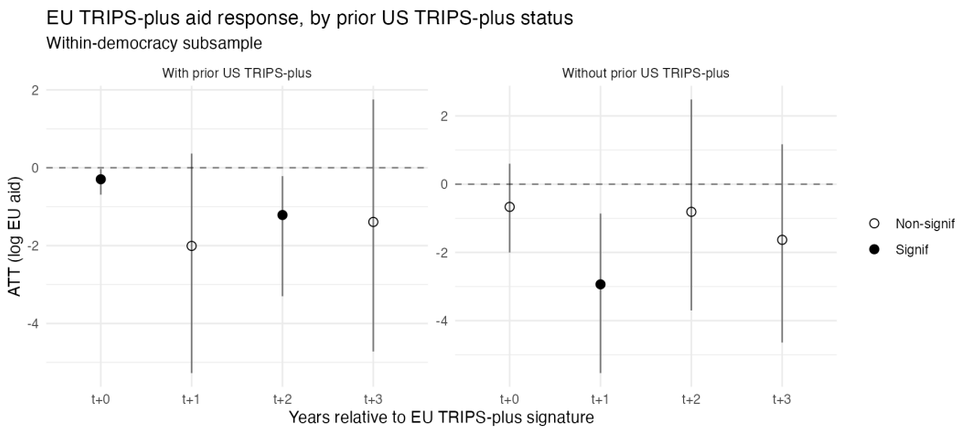


Figure 5: EU aid response to TRIPS-plus signature, split by prior US TRIPS-plus status. Within-democracy subsamples.

Alt-text. Two panels of EU within-democracy aid ATTs, split by whether the recipient already held a US TRIPS-plus PTA. Both panels show non-positive responses, with somewhat smaller magnitudes in the with-prior-US group.

Capacity building for the IP apparatus. The intellectual property apparatus is the machinery that administers the new obligations, meaning the patent office, its examiners, the registries, and the enforcement bodies. Assistance that builds it serves the rights holders, predominantly the donor's own firms, and strengthens enforcement of the very obligations the injured groups opposed. On the definition adopted in Section 2, compensation reaches the constituencies bearing the adjustment costs whatever fiscal route it takes, so apparatus assistance, though a transfer to the recipient state, sits on the far side of that line. If the signature-year increase consisted of such assistance, the compensation reading would fail and a donor-serving reading would take its place.

No CRS purpose code names intellectual property, so I remove from the outcome an envelope of codes that over-covers any such assistance. It comprises trade policy and administrative management (33110), whose scope includes implementation of multilateral trade agreements, trade facilitation (33120), which covers customs and other border agencies, regional trade agreements (33130), multilateral trade negotiations (33140), trade education and training (33181), business policy and administration (25010), whose scope includes property rights and competition policy, le-

gal and judicial development (15130), and small and medium-sized enterprise development (32130). Together these account for 4.3% of US disbursements over the coverage window.

With treatment, matched sets, and CBPS weights held identical, the $t + 0$ ATT on the remaining aid is 0.312 against 0.310 on the unrestricted series, so the apparatus envelope accounts for none of the surge.

Two limits attach. The exercise rules out the donor-serving reading without confirming the compensation reading, because the activity records identify purpose, delivery modality, and implementing partner without identifying who ultimately benefits. And earmarked apparatus money could free other resources and so compensate indirectly, though an envelope of 4.3% of disbursements leaves that channel little to work with. The same records place the recipient-government share of the signature-year transfer between 2% and 73%, as the footnote above sets out. Restricting the outcome to a single bucket locates where the response sits, and Table A.12 in the Appendix reports the full set of restricted-outcome estimates. The governance and general budget support codes give a $t + 0$ ATT of 1.121 and the trade and IP apparatus codes 3.290, both statistically indistinguishable from zero, with the apparatus reaching a significant 3.794 at $t + 3$. Both series sit at exactly zero in 18% and 32% of estimation-panel country-years respectively, so their coefficients register movement onto the extensive margin and carry no reading as a proportional change in dollars. By delivery modality, technical assistance returns a significant $t + 0$ ATT of 0.212 and project-type assistance 0.325, against 0.310 on the unrestricted series, so the response spreads across every channel of delivery. Crossing the two axes, the treated democracies received 272.7 million dollars carrying apparatus purpose codes over FY2003 through FY2010, 242.5 million of it as funded projects and 30.2 million as technical cooperation. The project cell carries a significant $t + 0$ ATT of 4.490 that persists at 5.015 for $t + 1$ and 5.627 for $t + 3$, while the technical cooperation cell sits at exactly zero in 59% of country-years and returns no significant lead. The apparatus money therefore arrives overwhelmingly as funded projects on the multi-year horizon on which institution building is funded, while total aid shows a single-year pulse. Delivery modality identifies neither the recipients of the funds nor their political incidence, so the split describes how the apparatus component is delivered and carries no inference about who is compensated.

Chinese development finance. The panel spans the rise of China as an alternative financier, which raises the possibility that the US surge reflects recipients hedging against Chinese influence.

Merging recipient-year Chinese official finance commitments from AidData (Dreher et al., 2022) shows the opposite. Seven of the nine treated democracies received no Chinese finance across the pre-treatment window, against a sample average in the hundreds of millions of dollars per year, since these are pre-BRI Latin American economies, several of which recognized Taiwan. Adding lagged Chinese finance to the covariate set leaves the $t + 0$ estimate unattenuated (Table A.13 in the Appendix).

4 Conclusion

TRIPS turned aspirational intellectual property norms into enforceable commitments, and the United States and the European Union have since pushed past that floor by adding TRIPS-plus provisions to successive preferential trade agreements. US chapters go substantially further than EU chapters, particularly on patent linkage, pharmaceutical data exclusivity, and enforcement. This study asks which of the developing economies that accept these rules are compensated for the concession and why donor responses diverge so sharply.

The answer sets out three conditions, two on the donor side and one on the recipient side. The donor's IP demands must be intrusive enough to generate politically salient adjustment costs, and its aid governance must be centralized enough to tie disbursements to a discrete policy concession. The United States satisfies both through its maximalist IP demands and its concentrated allocation authority, deploying aid and trade as joint instruments of donor strategy in the manner described by Bermeo (2017) and Bermeo (2018). The European Union satisfies neither, with IP chapters that are lighter on the provisions bearing most directly on generic entry and authority over development assistance fragmented across the Commission and twenty-seven member states in a way that anchors EU aid in developmental categories (Carbone, 2013; Dietrich, 2021). The third condition sits with the recipient, whose institutions must make the commitment credible to the donor, and regime type carries that condition in the results.

Democratic developing countries signing TRIPS-plus agreements with the United States receive an aid surge of roughly 36% in the year of signature and no further effect at later leads. The observed autocratic signatories of US agreements show no comparable increase, though each autocracy arm rests on three treated countries and is reported as a descriptive comparison. Signatories of EU TRIPS-plus agreements show no sustained positive response from either regime

type. Within the US arm the surge concentrates among democracies with prior major-donor PTA partners, consistent with the implementation-credibility condition, and the design reads it at the level of the agreement as a whole, whose political price the intellectual property chapter raises.

Work on the political economy of TRIPS-plus has traced how US industry coalitions carried high-standard IP rules into trade policy and how bilateral commitments then raised the baseline for later multilateral bargaining (Sell, 2003; Sell, 2010; Drahos and Braithwaite, 2002; Morin, 2009). Quantitative studies have mapped the depth of IP provisions across agreements and linked it to the power and innovative capacity of the parties (Morin and Surbeck, 2020; Dür and Mödlhamer, 2022), and on the developing-country side the literature has documented what the chapters cost in medicine prices and industrial policy space (Shadlen, 2005; R. D. Smith et al., 2009; Duggan et al., 2016). It has said less about whether the demanding party pays for the concession, and that is the question this paper takes up. The gap between the heavier US template and the lighter EU template, long documented in the text of the chapters (Fink and Reichenmiller, 2006; Roffe, 2014), reappears in the aid that follows signature. The organized losers this literature identifies, generic producers and patient groups prominent among them, supply the political demand that a donor transfer would answer, and the coalition that formed in Peru shows how far that demand reached beyond the pharmaceutical sector. With market-access leverage held as a background condition, aid appears as a second instrument in TRIPS-plus bargaining, and whether a donor can use it depends on how its own aid institutions are organized.

In the aid literature, formal accounts model aid as a purchase of policy concessions from recipient leaders (Bueno de Mesquita and A. Smith, 2009; Bueno de Mesquita and A. Smith, 2007), and empirical studies find aid moving with recipients' votes in the UN General Assembly and with their entry into preferential trade agreements (Dreher et al., 2008; Carter and Stone, 2015; Baccini and Urpelainen, 2012). These accounts treat the donor as a single principal able to time a payment to a concession, and the results identify that capacity as a condition for the exchange. The European Union lacks it, and despite its market leverage its aid shows no sustained response to TRIPS-plus signature. The contrast ties aid-for-policy bargaining to work on how donor institutions shape the strategic use of aid (Dietrich, 2021; Kim and Jensen, 2018), though the design moves intrusiveness and allocation authority together across donors and so reads the two donor-side conditions jointly.

The TRIPS-plus case also widens the set of concessions that aid has been found to accompany. A UN vote is a single act, while a TRIPS-plus chapter commits the recipient to regulatory

changes whose costs arrive over years of implementation. The credibility a donor needs therefore concerns how long the commitment holds, and the canonical worry that leaders will divert fungible aid matters less (Bueno de Mesquita and A. Smith, 2009). The compensation logic holds whatever route the money takes, which fits evidence that US assistance often bypasses recipient budgets (Dietrich, 2013). Among the observed US signatories the response is confined to democracies, consistent with work that places the binding constraint on US aid in congressional politics (Carter and Stone, 2015; Milner and Tingley, 2010). The case attaches a specific concession and a specific moment of commitment to accounts of trade and aid as joint instruments of donor strategy (Bermeo, 2017; Bermeo, 2018; Bermeo and Leblang, 2015).

Research on PTA design has mapped the spread of regulatory provisions into trade agreements (Dür et al., 2014) and traced their depth to the investment interests of firms, the market power of the larger partner, and the preferences of domestic interest groups (Manger, 2009; Manger and Shadlen, 2014; Dür et al., 2023). Rodrik (2018) argues that much of this regulatory content, intellectual property rules prominent among it, reflects rent-seeking by well-connected firms in the demanding countries. Work on side payments treats aid as support for the agreement as a whole (Baccini and Urpelainen, 2012). The results here are consistent with a single intrusive chapter raising the political price of a deep agreement, since EU agreements impose agricultural and services adjustment on a scale similar to US agreements and draw no comparable response. The design reads the response at the level of the agreement, so the attribution to the IP chapter remains an inference. If it holds, side payments of this kind answer in part to the content of particular chapters, and measures of PTA design drawn from treaty texts capture only part of the exchange through which deep provisions are accepted. The credibility pattern connects to work on PTAs as commitment devices for reforming governments (Baccini and Urpelainen, 2014). The US response concentrated among recipients that had already carried a major-donor PTA, which suggests that a partner's record under earlier agreements may shape the terms of the next, though the split cannot separate that record from income or administrative capacity.

The mechanism also carries a policy implication. As the institutional foundations of US development assistance come under strain and its allocation machinery is reorganized, a donor that loses the ability to deliver targeted transfers loses the instrument that made intrusive IP demands politically absorbable for its partners.

The theory also suggests implications for adoption and for the depth of commitments,

neither of which is an outcome in this design, and they are offered as directions for further work. Deep regulatory chapters should travel more easily when the demanding party can pair them with side payments, so standards would diffuse unevenly, advancing where a donor can both demand and compensate and stalling where the demand outruns the capacity to deliver, and the reach of TRIPS-plus norms would depend less on their normative appeal than on the political economy of the donors that carry them. The same logic suggests one reason the United States pressed its post-TRIPS agenda through bilateral agreements (Sell, 2010; Morin, 2009), since a bilateral venue lets a demanding party pair a regulatory demand with a transfer aimed at one partner at the moment it signs. What this design establishes is narrower. Among developing countries accepting TRIPS-plus obligations, a signature-year aid response appears for democratic signatories of US agreements and for no other combination of donor and regime observed here.

References

Alesina, Alberto and David Dollar (2000). "Who Gives Foreign Aid to Whom and Why?" In: *Journal of Economic Growth* 5.1, pp. 33–63.

Antoine, Elise, Ece Özlem Atikcan, and Adam William Chalmers (2024). "Politicisation, business lobbying, and the design of preferential trade agreements". In: *Journal of European Public Policy* 31.1, pp. 239–268.

Arase, David (1995). *Buying Power: The Political Economy of Japan's Foreign Aid*. Boulder, CO: Lynne Rienner Publishers.

Baccini, Leonardo and Johannes Urpelainen (2012). "Strategic side payments: Preferential trading agreements, economic reform, and foreign aid". In: *The Journal of Politics* 74.4, pp. 932–949.

— (2014). "International institutions and domestic politics: can preferential trading agreements help leaders promote economic reform?" In: *The Journal of Politics* 76.1, pp. 195–214.

Bearce, David H. and Daniel C. Tirone (2010). "Foreign aid effectiveness and the strategic goals of donor governments". In: *Journal of Politics* 72.3, pp. 837–851.

Bermeo, Sarah Blodgett (2017). "Aid Allocation and Targeted Development in an Increasingly Connected World". In: *International Organization* 71.4, pp. 735–766.

— (2018). *Targeted development: Industrialized country strategy in a globalizing world*. Oxford University Press.

Bermeo, Sarah Blodgett and David Leblang (2015). "Migration and Foreign Aid". In: *International Organization* 69.3, pp. 627–657.

Branstetter, Lee G., Raymond Fisman, and C. Fritz Foley (2006). "Do Stronger Intellectual Property Rights Increase International Technology Transfer? Empirical Evidence from U.S. Firm-Level Panel Data". In: *Quarterly Journal of Economics* 121.1, pp. 321–349.

Bueno de Mesquita, Bruce and Alastair Smith (2007). "Foreign aid and policy concessions". In: *Journal of Conflict Resolution* 51.2, pp. 251–284. ISSN: 00220027.

— (2009). "A Political Economy of Aid". In: *International Organization* 63.2, pp. 309–340.

Carbone, Maurizio (2007). *The European Union and international development: The politics of foreign aid*. Routledge.

— (2012). "Beyond Aid: Policy Coherence and Europe's Development Policy". In: *International Development Policy* 3, pp. 161–173.

— (2013). "Between EU Actorness and Aid Effectiveness: The Logics of EU Aid to Sub-Saharan Africa". In: *International Relations* 27.3, pp. 341–355.

Carter, David B and Randall W Stone (2015). "Democracy and multilateralism: the case of vote buying in the UN General Assembly". In: *International Organization* 69.1, pp. 1–33.

Coppedge, Michael et al. (2026). *V-Dem [Country-Year/Country-Date] Dataset v16*. Varieties of Democracy (V-Dem) Project.

Dietrich, Simone (2013). "Bypass or Engage? Explaining Donor Delivery Tactics in Foreign Aid Allocation". In: *International Studies Quarterly* 57.4, pp. 698–712.

— (2021). *States, Markets, and Foreign Aid*. Cambridge University Press.

Dollar, David and Victoria Levin (2006). "The Increasing Selectivity of Foreign Aid, 1984–2003". In: *World Development* 34.12, pp. 2034–2046.

Drahos, Peter and John Braithwaite (2002). *Information Feudalism: Who Owns the Knowledge Economy?* London: Earthscan.

Dreher, Axel, Peter Nunnenkamp, and Rainer Thiele (2008). "Does US aid buy UN general assembly votes? A disaggregated analysis". In: *Public Choice* 136.1-2, pp. 139–164. ISSN: 00485829.

Dreher, Axel, Jan-Egbert Sturm, and James Raymond Vreeland (2009). "Development aid and international politics: Does membership on the UN Security Council influence World Bank decisions?" In: *Journal of Development Economics* 88.1, pp. 1–18.

Dreher, Axel et al. (2022). *Banking on Beijing: The Aims and Impacts of China's Overseas Development Program*. Cambridge: Cambridge University Press. ISBN: 978-1-108-47410-8.

Drexl, Josef (2014). "Intellectual Property and Implementation of Recent Bilateral Trade Agreements in the EU". In: *EU Bilateral Trade Agreements and Intellectual Property: For Better or Worse?* MPI Studies on Intellectual Property and Competition Law. Springer, pp. 265–291.

Duggan, Mark, Craig Garthwaite, and Aparajita Goyal (2016). "The Market Impacts of Pharmaceutical Product Patents in Developing Countries: Evidence from India". In: *American Economic Review* 106.1, pp. 99–135.

Dür, Andreas, Leonardo Baccini, and Manfred Elsig (2014). "The design of international trade agreements: Introducing a new dataset". In: *The Review of International Organizations* 9.3, pp. 353–375.

Dür, Andreas and Christoph Mödlhamer (2022). "Power and innovative capacity: Explaining variation in intellectual property rights regulation across trade agreements". In: *International Interactions* 48.1, pp. 23–48.

Dür, Andreas et al. (2023). "Interest group preferences towards trade agreements: Institutional design matters". In: *Interest Groups & Advocacy* 12.1, pp. 48–72.

Farrell, Henry and Abraham L Newman (2019). "Weaponized interdependence: How global economic networks shape state coercion". In: *International Security* 44.1, pp. 42–79.

Fink, Carsten and Patrick Reichenmiller (2006). "Tightening TRIPS: Intellectual property provisions of US free trade agreements". In: *Trade, Doha, and Development: A Window into the Issues*. Ed. by Richard Newfarmer. Washington, DC: World Bank, pp. 289–303.

Fleck, Robert K. and Christopher Kilby (2010). "Changing Aid Regimes? U.S. Foreign Aid from the Cold War to the War on Terror". In: *Journal of Development Economics* 91.2, pp. 185–197.

García, Luis Alonso (2008). *Intellectual Property, Bilateral Agreements and Sustainable Development: Intellectual Property in the US-Peru Trade Promotion Agreement*. Intellectual Property, Bilateral Agreements and Sustainable Development Series 3. Center for International Environmental Law.

Glass, Amy Jocelyn and Xiaodong Wu (2007). "Intellectual property rights and quality improvement". In: *Journal of Development Economics* 82.2, pp. 393–415.

Helfer, Laurence R. (2004). "Regime Shifting: The TRIPs Agreement and New Dynamics of International Intellectual Property Lawmaking". In: *Yale Journal of International Law* 29.1, pp. 1–83.

Imai, Kosuke, In Song Kim, and Erik H Wang (2023). "Matching methods for causal inference with time-series cross-sectional data". In: *American Journal of Political Science* 67.3, pp. 587–605.

Kilby, Christopher (2009). "The Political Economy of Conditionality: An Empirical Analysis of World Bank Loan Disbursements". In: *Journal of Development Economics* 89.1, pp. 51–61.

Kim, Youngwan and Christian Jensen (2018). "Preferences and institutions: constraints on European Union foreign aid distribution". In: *Journal of European Integration* 40.2, pp. 177–192. ISSN: 14772280.

Knack, Stephen and Aminur Rahman (2007). "Donor Fragmentation and Bureaucratic Quality in Aid Recipients". In: *Journal of Development Economics* 83.1, pp. 176–197.

Lancaster, Carol (2006). *Foreign Aid: Diplomacy, Development, Domestic Politics*. Chicago: University of Chicago Press.

Langton, Danielle (2008). *Trade Capacity Building: Foreign Assistance for Trade and Development*. CRS Report RL33628. Congressional Research Service.

Lawson, Marian L. (2016). *Foreign Aid: An Introduction to U.S. Programs and Policy*. CRS Report R40213. Congressional Research Service.

Lindstrom, Beatrice (2010). "Scaling back TRIPS-plus: An analysis of intellectual property provisions in trade agreements and implications for Asia and the Pacific". In: *NYUJ Int'l L. & Pol.* 42.3, pp. 917–980.

Manger, Mark S (2009). *Investing in protection: The politics of preferential trade agreements between north and south*. Cambridge University Press.

Manger, Mark S and Kenneth C Shadlen (2014). "Political trade dependence and North–South trade agreements". In: *International studies quarterly* 58.1, pp. 79–91.

Mangini, Michael-David (2026). "How effective is trade conditionality? Economic coercion in the Generalized System of Preferences". In: *The Review of International Organizations* 21.1, pp. 143–162.

Mansfield, Edward D, Helen V Milner, and B Peter Rosendorff (2000). "Free to trade: Democracies, autocracies, and international trade". In: *American political science review* 94.2, pp. 305–321.

Marshall, Monty G. and Ted Robert Gurr (2020). *Polity V: Political Regime Characteristics and Transitions, 1800-2018*. Center for Systemic Peace.

Martin, Lisa L (2000). *Democratic commitments: Legislatures and international cooperation*. Princeton, NJ: Princeton University Press.

Maskus, Keith E. (2000). "Intellectual Property Rights and Economic Development". In: *Case Western Reserve Journal of International Law* 32.3, pp. 471–506.

— (2014). "Assessing the Development Promise of IP Provisions in EU Economic Partnership Agreements". In: *EU Bilateral Trade Agreements and Intellectual Property: For Better or Worse?* Springer, pp. 171–188.

McCalman, Phillip (2001). "Reaping what you sow: an empirical analysis of international patent harmonization". In: *Journal of International Economics* 55.1, pp. 161–186.

Milner, Helen V and Dustin H Tingley (2010). "The political economy of US foreign aid: American legislators and the domestic politics of aid". In: *Economics & Politics* 22.2, pp. 200–232.

Morin, Jean-Frédéric (2009). "Multilateralizing TRIPs-Plus Agreements: Is the US Strategy a Failure?" In: *The Journal of World Intellectual Property* 12.3, pp. 175–197.

Morin, Jean-Frédéric and Jenny Surbeck (2020). "Mapping the New Frontier of International IP Law: Introducing a TRIPs-plus Dataset". In: *World Trade Review* 19.1, pp. 109–122.

Office of the United States Trade Representative (2005). *Peru TPA Facts*. Tech. rep. Office of the United States Trade Representative.

Öhler, Hannes, Peter Nunnenkamp, and Axel Dreher (2012). "Does Conditionality Work? A Test for an Innovative US Aid Scheme". In: *European Economic Review* 56.1, pp. 138–153.

Park, Jihye (2026). "Aid, lending, and TRIPS". In: *The Review of International Organizations*. Advance online publication.

Rodrik, Dani (2018). "What do trade agreements really do?" In: *Journal of Economic Perspectives* 32.2, pp. 73–90.

Roffe, Pedro (2014). "Intellectual Property Chapters in Free Trade Agreements: Their Significance and Systemic Implications". In: *EU Bilateral Trade Agreements and Intellectual*

Property: For Better or Worse? MPI Studies on Intellectual Property and Competition Law. Springer, pp. 17–40.

Sell, Susan K. (2003). *Private Power, Public Law: The Globalization of Intellectual Property Rights*. Cambridge: Cambridge University Press.

— (2010). "The Rise and Rule of a Trade-Based Strategy: Historical Institutionalism and the International Regulation of Intellectual Property". In: *Review of International Political Economy* 17.4, pp. 762–790.

Seuba, Xavier (2014). "Implementation Issues Arising from Intellectual Property Chapters Contained in Trade Agreements Between the EU and Developing Countries". In: *EU Bilateral Trade Agreements and Intellectual Property: For Better or Worse?* Springer, pp. 293–303.

Shadlen, Kenneth C (2005). "Exchanging development for market access? Deep integration and industrial policy under multilateral and regional-bilateral trade agreements". In: *Review of international political economy* 12.5, pp. 750–775.

Shadlen, Kenneth C et al. (2011). "Globalization, Intellectual Property Rights, and Pharmaceuticals: Meeting the Challenges to Addressing Health Gaps in the New International Environment". In: *Intellectual Property, Pharmaceuticals and Public Health*. Edward Elgar.

Simmons, Beth A (2000). "International Law and State Behavior: Commitment and Compliance in International Monetary Affairs". In: *American Political Science Review* 94.4, pp. 819–835.

Simmons, Beth A, Frank Dobbin, and Geoffrey Garrett (2006). "Introduction: The International Diffusion of Liberalism". In: *International Organization* 60.4, pp. 781–810.

Smith, Richard D, Carlos Correa, and Cecilia Oh (2009). "Trade, TRIPS, and pharmaceuticals". In: *The Lancet* 373.9664, pp. 684–691.

Turk, Matthew (2010). "Bargaining and Intellectual Property Treaties: The Case for a Pro-Development Interpretation of TRIPS but Not TRIPS Plus". In: *NYUJ Int'l L. & Pol.* 42.3, p. 981.

A Appendix

Contents

A.1 Comparison of TRIPS-Plus Provisions in US and EU Trade Agreements 35
A.2 Descriptive statistics - EU aid recipients 36
A.3 Descriptive statistics - US aid recipients 36
A.4 PanelMatch ATTs by donor and recipient regime type 36
A.5 Pre-treatment placebo ATTs at $t-4$, $t-3$, $t-2$ 36
A.6 US within-democracy ATTs split by prior major-donor PTA experience 37
A.7 EU aid response to non-TRIPS-plus PTA signatures 37
A.8 EU MFN sequencing: within-democracy ATTs split by prior US TRIPS-plus status . 37
A.9 Treated cases and matched sets by research arm 38
A.10 Robustness of the four-arm $t+0$ ATTs to alternative democracy codings 38
A.11 Influence diagnostics for the US within-democracy TRIPS-plus effect 38
A.12 Capacity-building apparatus: PanelMatch ATTs on restricted US aid outcomes . . . 39
A.13 US within-democracy $t+0$ ATT with and without Chinese finance controls 39

Table A.1: Comparison of TRIPS-Plus Provisions in US and EU Trade Agreements

Provision	**United States**	**European Union**
Patent Protection		
Patent term extension	Required; compensates for regulatory and patent office delays (typically 3–5 years additional)	Required via Supplementary Protection Certificates (SPCs); up to 5 years extension
Patentable subject matter	Broad scope; includes plants, animals, and surgical methods in some FTAs	Generally narrower; excludes plants and animals consistent with TRIPS flexibilities
Second-use patents	Explicitly permitted in most FTAs	Not typically required
Compulsory licensing	Restricted to national emergencies, anti-competitive remedies, and public non-commercial use (pre-2007 FTAs); flexibilities restored in post-May 2007 FTAs	Generally preserves TRIPS flexibilities; fewer explicit restrictions on compulsory licensing grounds
Patent linkage	Required; regulatory authorities must notify patent holders before approving generics	Not required; EU relies on judicial enforcement rather than regulatory linkage
Data Exclusivity		
Pharmaceutical test data	5 years for new chemical entities; 3 years for new clinical information	8 years data exclusivity + 2 years market exclusivity + 1 year for new indication (8+2+1 formula)
Biologics	No biologics-specific exclusivity in bilateral FTAs with developing countries; standard 5-year NCE provisions apply	Covered under same 8+2+1 framework as small molecules
Protection against reliance	Prohibits reliance on originator data for generic approval during exclusivity period	Similar prohibition; additionally bars generic marketing for 2 years after data exclusivity expires
Other Key Provisions		
Geographical indications	Opposes expansion beyond wines and spirits; prefers trademark-based protection	Strongly promotes GI protection for agricultural and non-agricultural products; requires partner countries to protect listed EU GIs
Enforcement	Extensive provisions: criminal penalties for IP infringement, border measures, statutory damages, *ex officio* enforcement	Criminal enforcement for commercial-scale infringement; emphasis on civil and border measures
Exhaustion of rights	Generally supports national exhaustion (restricting parallel imports) in FTAs	More flexible; some agreements silent on exhaustion, preserving policy space
Traditional knowledge	Limited provisions	Increasingly includes provisions on genetic resources and traditional knowledge disclosure

Note: US provisions reflect the standard FTA template, though post-May 2007 agreements (Peru, Colombia, Panama) include modified pharmaceutical provisions following the bipartisan trade deal. EU provisions reflect the standard Association Agreement and Economic Partnership Agreement templates. Both donors' provisions exceed the TRIPS Agreement minimum standards ("TRIPS-plus"), but differ in emphasis: the US prioritizes patent linkage, enforcement, and restrictions on compulsory licensing, while the EU emphasizes longer data exclusivity periods and geographical indication protection.

Table A.2: Descriptive statistics - EU aid recipients

Treatment and outcome	Mean	SD	Min	Max	*N*
TRIPS-plus signed (dummy)	0.015	0.120	0	1	2,387
EU aid (logged)	16.35	7.377	-20.08	22.52	2,112
Matching covariates	Mean	SD	Min	Max	*N*
GDP per capita (logged)	8.934	0.998	6.752	11.784	2,316
GDP (logged)	24.63	2.048	20.16	30.56	2,316
GDP growth (%)	4.009	5.123	-36.392	63.440	2,365
Democracy	0.658	0.474	0	1	2,387

Table A.3: Descriptive statistics - US aid recipients

Treatment and outcome	Mean	SD	Min	Max	*N*
TRIPS-plus signed (dummy)	0.005	0.100	0	1	3,150
US aid (logged)	17.25	2.599	-15.74	23.48	2,498
Matching covariates	Mean	SD	Min	Max	*N*
GDP per capita (logged)	8.833	0.966	6.289	10.960	2,960
GDP (logged)	24.38	2.002	20.01	30.20	2,960
GDP growth (%)	3.817	5.412	-36.392	106.280	3,001
Democracy	0.696	0.460	0	1	2,760

Table A.4: PanelMatch ATTs by donor and recipient regime type

	EU aid		US aid	
	Democracies	Autocracies	Democracies	Autocracies
	(1)	(2)	(3)	(4)
$t+0$	-0.587	0.519	0.309^{+}	-0.502
	(0.599)	(0.306)	(0.181)	(0.389)
$t+1$	-2.575^{**}	0.074	-0.064	-0.632
	(0.989)	(0.227)	(0.150)	(0.490)
$t+2$	-0.897	-1.231^{*}	0.078	-1.542^{*}
	(1.045)	(0.568)	(0.161)	(1.104)
$t+3$	-1.253	-0.082	0.170	-1.040
	(1.329)	(0.553)	(0.221)	(1.244)
Treated unit-years	385	19	183	86

Bootstrapped standard errors in parentheses (1,000 iterations)
*Signif. codes: **: 0.01, *: 0.05, +: 0.1*

Table A.5: Pre-treatment placebo ATTs at $t-4$, $t-3$, $t-2$

	EU aid		US aid	
	Democracies	Autocracies	Democracies	Autocracies
	(1)	(2)	(3)	(4)
$t-4$	-0.168	1.087^{*}	-0.250	-1.079^{**}
	(0.597)	(0.477)	(0.426)	(0.145)
$t-3$	-0.221	0.371	-0.230	-0.877^{**}
	(0.872)	(0.325)	(0.361)	(0.259)
$t-2$	0.074	-0.146	-0.165	-0.068
	(0.550)	(0.524)	(0.546)	(0.121)
Treated unit-years	385	19	183	86

Bootstrapped standard errors in parentheses (1,000 iterations)
*Signif. codes: **: 0.01, *: 0.05, +: 0.1*

Table A.6: US within-democracy ATTs split by prior major-donor PTA experience

	With prior major-donor PTA (1)	Without prior major-donor PTA (2)
$t+0$	0.586**	0.090
	(0.220)	(0.259)
$t+1$	0.156	-0.154
	(0.268)	(0.227)
$t+2$	0.317+	-0.241
	(0.240)	(0.275)
$t+3$	0.425+	-0.301
	(0.265)	(0.270)
Treated unit-years	102	81

Bootstrapped standard errors in parentheses (1,000 iterations)
Major-donor PTAs include US, EU and member states, EFTA, Canada, Japan, Korea, Australia, New Zealand
*Signif. codes: **: 0.01, *: 0.05, +: 0.1*

Table A.7: EU aid response to non-TRIPS-plus PTA signatures

	Pooled (1)	Within-democracy (2)
$t+0$	0.566	0.743
	(0.746)	(1.250)
$t+1$	1.664**	2.002**
	(0.889)	(1.470)
$t+2$	1.160*	1.658**
	(0.860)	(1.570)
$t+3$	1.304*	0.924
	(1.112)	(1.520)
Treated unit-years	128	108

Bootstrapped standard errors in parentheses (1,000 iterations)
*Signif. codes: **: 0.01, *: 0.05, +: 0.1*

Table A.8: EU MFN sequencing: within-democracy ATTs split by prior US TRIPS-plus status

	With prior US TRIPS-plus (1)	Without prior US TRIPS-plus (2)
$t+0$	-0.295+	-0.666
	(0.214)	(0.811)
$t+1$	-2.008	-2.935**
	(1.878)	(1.427)
$t+2$	-1.214*	-0.810
	(1.056)	(1.918)
$t+3$	-1.391	-1.631
	(2.059)	(1.783)
Treated unit-years	95	290

Bootstrapped standard errors in parentheses (1,000 iterations)
*Signif. codes: **: 0.01, *: 0.05, +: 0.1*

Table A.9: Treated cases and matched sets by research arm

Arm	Treated countries	Treated country-years	Matched sets (non-empty)	Mean matched-set size	Control country-years
US, democracy	9	9	9	98.0	393
US, autocracy	3	3	3	101.0	202
EU, democracy	31	32	31	73.0	393
EU, autocracy	3	3	3	85.7	257

Note: Treated country-years are treatment onsets usable by PanelMatch with a four-year lag window. Matched sets are refined with CBPS weighting. Control country-years count unique control country-year pairs across non-empty matched sets.

Table A.10: Robustness of the four-arm $t + 0$ ATTs to alternative democracy codings

	EU		US	
Democracy measure	Democracies	Autocracies	Democracies	Autocracies
Polity2 > 0 (baseline)	-0.587	0.519	0.309^{+}	-0.502
	(0.599)	(0.306)	(0.181)	(0.389)
Polity2 ≥ 6	-0.720	1.562	0.309^{+}	-0.502
	(0.706)	(1.622)	(0.181)	(0.389)
V-Dem polyarchy ≥ 0.5	-1.019	0.463^{*}	0.309^{+}	-0.389
	(0.744)	(0.360)	(0.181)	(0.283)

Note: $t + 0$ ATT in log aid units with bootstrapped standard errors in parentheses (1,000 iterations). Signif. codes: **: 0.01, *: 0.05, +: 0.1. Treated country-years per cell, ordered EU democracies, EU autocracies, US democracies, US autocracies, are baseline 32/3/9/3, Polity2$\geq$6 28/12/9/3, and V-Dem 24/9/9/5. The US within-democracy estimate is numerically identical across all three codings because every democratic US signatory clears each threshold in its signature year.

Table A.11: Influence diagnostics for the US within-democracy TRIPS-plus effect

Sample	$t + 0$ ATT	Bootstrapped SE	Refinement
Full sample	0.309^{+}	(0.181)	CBPS.weight
Drop Chile	0.311^{+}	(0.206)	CBPS.weight
Drop Colombia	0.244	(0.193)	CBPS.weight
Drop Costa Rica	0.313	(0.192)	CBPS.weight
Drop Dominican Republic	0.302	(0.195)	CBPS.weight
Drop El Salvador	0.273	(0.205)	CBPS.weight
Drop Guatemala	0.294	(0.184)	CBPS.weight
Drop Honduras	0.337^{+}	(0.196)	CBPS.weight
Drop Panama	0.426^{*}	(0.207)	CBPS.weight
Drop Peru	0.232	(0.170)	CBPS.weight
Drop reform-credential five	0.081	(0.240)	CBPS.weight
Joint exclusion, dynamic effects			
t+0	0.081	(0.240)	CBPS.weight
t+1	-0.134	(0.273)	CBPS.weight
t+2	-0.047	(0.269)	CBPS.weight
t+3	-0.155	(0.238)	CBPS.weight

Each row re-estimates the canonical PanelMatch specification (lag 4, CBPS-weight refinement, leads 0–3, 1,000 block-bootstrap iterations) after excluding the listed country entirely. The joint exclusion drops Chile, Colombia, Costa Rica, the Dominican Republic, and Peru. Bootstrapped standard errors in parentheses. Signif. codes: **: 0.01, *: 0.05, +: 0.1.

Table A.12: Capacity-building apparatus: PanelMatch ATTs on restricted US aid outcomes

	$t+0$	$t+1$	$t+2$	$t+3$
A. Removing an over-inclusive apparatus envelope from the outcome				
Total aid (activity file)	0.310^{+}	-0.059	0.084	0.175
	(0.180)	(0.149)	(0.161)	(0.221)
Aid excluding apparatus codes	0.312^{+}	-0.049	0.068	0.152
	(0.182)	(0.150)	(0.162)	(0.220)
Aid excluding apparatus and narcotics	0.190^{+}	-0.041	0.115	0.182
	(0.118)	(0.165)	(0.173)	(0.223)
B. Restricting the outcome to a purpose bucket				
Governance and budget support	1.121	1.146	2.441	2.535
	(1.440)	(1.369)	(1.884)	(2.045)
Trade and IP apparatus	3.290	3.027	3.143	3.794^{+}
	(2.821)	(2.536)	(2.647)	(2.141)
C. Restricting the outcome to a delivery modality				
Technical assistance	0.212^{+}	0.122	0.069	0.043
	(0.114)	(0.148)	(0.162)	(0.174)
Project-type	0.325	-0.233	-0.046	0.145
	(0.242)	(0.199)	(0.191)	(0.288)
D. Apparatus purpose crossed with delivery modality				
Apparatus × technical assistance	0.139	-1.243	4.347	2.958
	(4.693)	(4.928)	(3.321)	(2.670)
Apparatus × project-type	4.490^{+}	5.015^{+}	3.812	5.627^{+}
	(2.186)	(2.364)	(2.437)	(1.982)

Each cell re-estimates the canonical US within-democracy specification (lag 4, CBPS-weight refinement, leads 0–3, 1,000 block-bootstrap iterations) with the outcome restricted as labeled, holding the treatment, matched sets, and covariate weights fixed. Outcomes use the sign-preserving log $\text{sign}(x)\log(1+|x|)$. All arms rest on the 183 treated unit-years of the US within-democracy arm. The apparatus envelope comprises OECD purpose codes 33110, 33120, 33130, 33140, 33181, 25010, 32130, and 15130 (4.3% of US disbursements). Governance comprises the 151xx codes less 15130 plus general budget support 51010. Governance sits at zero in 18% of estimation-panel country-years, the apparatus in 32%, and apparatus technical cooperation in 59%, so coefficients on those series register movement onto the extensive margin and carry no reading as a proportional change in dollars. Over FY2003 through FY2010 the treated democracies received $272.7 million carrying apparatus purpose codes, $242.5 million as funded projects and $30.2 million as technical cooperation. Bootstrapped standard errors in parentheses. Signif.: + marks a bootstrap percentile interval that excludes zero (90% level).

Table A.13: US within-democracy $t+0$ ATT with and without Chinese finance controls

Lead	Baseline	+ Chinese finance lags
$t+0$	0.309^{+}	0.855
	(0.181)	(1.114)
$t+1$	-0.064	0.502
	(0.150)	(1.101)
$t+2$	0.078	1.007
	(0.161)	(1.733)
$t+3$	0.170	0.915
	(0.221)	(1.622)
SMD lag 1–3 log(1 + Chinese finance), treated vs CBPS-weighted controls: -1.343		
Treated window mean 0.5 vs sample mean 449.0 (mUSD 2021/yr)		

Note: Point estimates in log aid units with bootstrapped standard errors in parentheses (1,000 iterations). Signif. codes: **: 0.01, *: 0.05, +: 0.1. Chinese finance is AidData GCDF 3.0 official finance commitments (Dreher et al., 2022). Treated democracies received almost no Chinese finance in the signature window, so the augmented specification is imprecise despite an unattenuated point estimate.